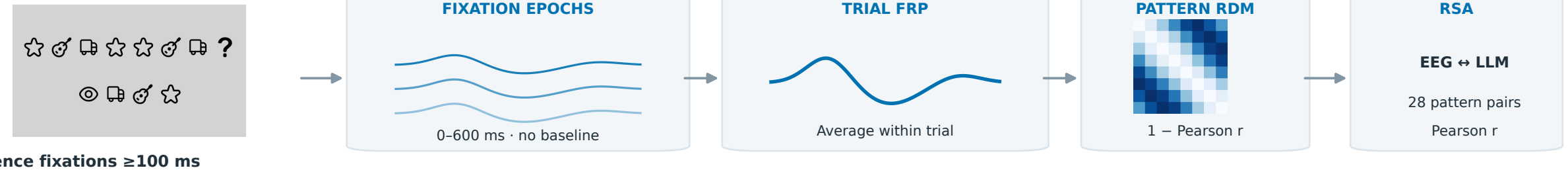
Abstract pattern reasoning in humans and language models

A common task links behavior, model representations and fixation-locked EEG

A From gaze fixations to representational similarity



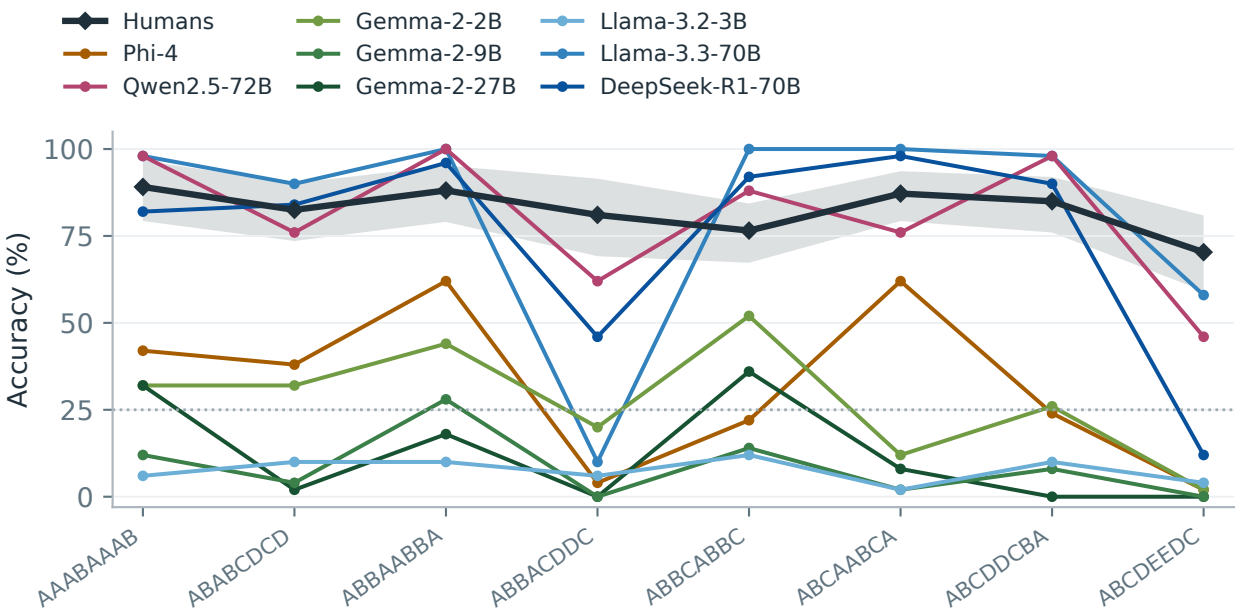
Sequence fixations ≥100 ms

8 patterns · 400 task items

Regional traces → pattern means / session → session RDMs → participant → group

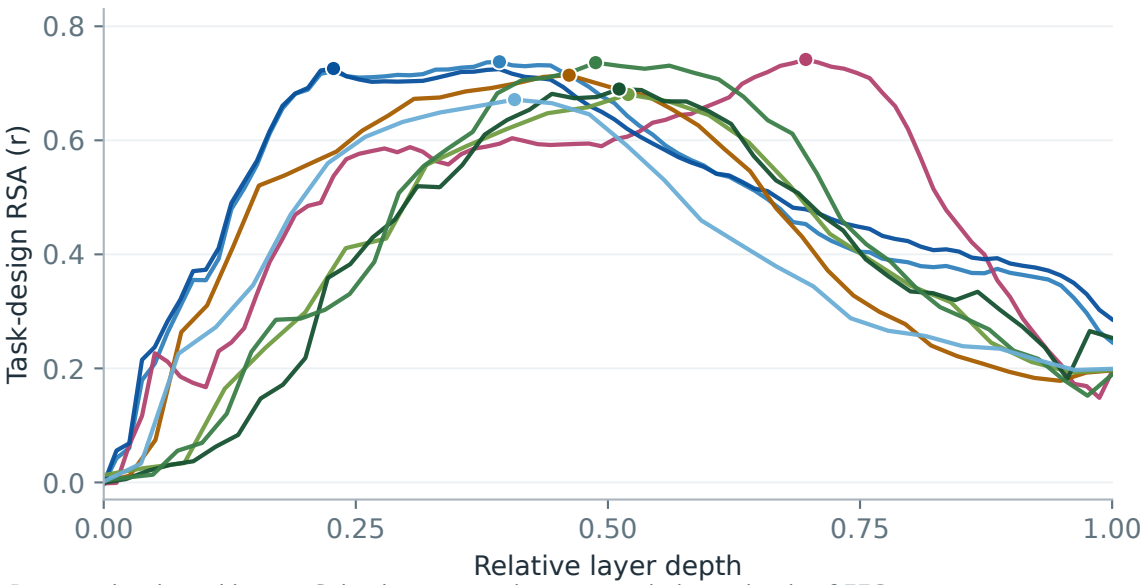
Schematic waveforms and RDM; LLM layers selected using task structure.

B Accuracy across abstract patterns



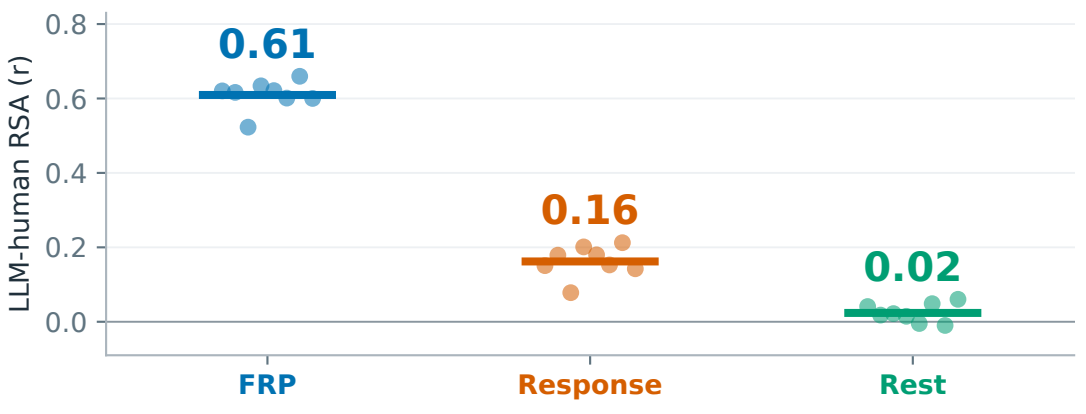
Human band: 95% bootstrap CI (N = 25). Dotted line: 25% chance.

C Task structure emerges inside the models



Dots mark selected layers. Selection uses task structure, independently of EEG.

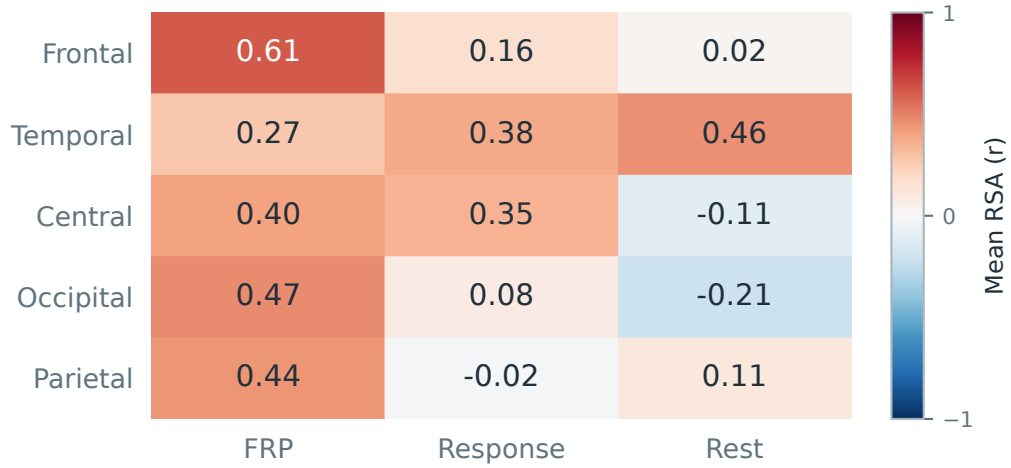
D Frontal correspondence is strongest for FRP



Dots: eight models. Horizontal marks: means. Matched N = 24.

Frontal FRP pattern-label test: p = .010; Holm-adjusted p = .030.

E Regional context matters



Means across eight models; color describes effect size, not significance.